A closed-form bound on metric inflation in windowed forecast evaluation

And a corrected, powered EEG forecasting benchmark

Kahlus / Neural-CASP

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Repository `Kahlus-V1`, branches `finding/overlap-inflation-law`, `finding/forecast-crossover-f8`

Scope. Two results: an exact bound on how much a window-scored forecast metric can overstate forecast skill — derived, not fitted, and violated by no (W, H) pair on any trace we tested — and a corrected benchmark that survived eight falsification tests, one of which broke a secondary claim we had made. That failure is reported here rather than removed. Every number is traceable to a named artifact.

1 The problem

A forecasting task is only a forecast if the target does not contain the input. The construction below is common in the biosignal literature and violates that silently. From a sequence of length $W+H$:

$$X = \text{seq}[0\!:\!W], \qquad Y = \text{seq}[H\!:\!H\!+\!W].$$

Then $Y[k] = X[k+H]$ whenever $k < W-H$: of the W scored target positions, $W-H$ are **copyable** — literally present in the input at a shifted index — and only H are strictly future. Averaging over all W positions therefore dilutes a genuine forecast error with positions a model can copy.

This is not the leakage the field guards against. It survives subject-wise partitioning, nested cross-validation and non-overlapping strides, because it lives *inside a single training example* rather than across the train/test split. We could not find it addressed in the leakage literature (Bernett et al., *Nat. Methods* 2024; Del Pup et al. 2025, five CV schemes over 10^5 trained EEG models; Apicella et al. 2024), all of which treat information shared *between* examples.

2 The law

Score **one model's one set of predictions** under three masks — all positions, the copyable prefix, the strictly-future tail. No argument about capacity, training or tuning can explain a gap between them. By construction,

$$\text{MSE}_{\text{overlap}} = \frac{(W-H)\,\text{MSE}_{\text{copy}} + H\,\text{MSE}_{\text{future}}}{W},$$

an identity rather than a fit, which rearranges to

$$\boxed{\text{inflation} \;\equiv\; \frac{\text{MSE}_{\text{future}}}{\text{MSE}_{\text{overlap}}} \;=\; \frac{W/H}{1 + \frac{W-H}{H} \cdot \frac{\text{MSE}_{\text{copy}}}{\text{MSE}_{\text{future}}}} \;\le\; \frac{W}{H}.}$$

Equality holds as $\text{MSE}_{\text{copy}} \to 0$. The bound is one-sided: a real result can never exceed it, so W/H is always safe as a worst case.

W/H **requires no data** — only the window length and horizon printed in a methods section. A paper reporting a 128-sample target window at horizon 1 is reporting a number that may be up to 128× better than its forecast error.

3 Verification

A 26-point grid, $W \in \{32, 64, 128, 256\} \times H \in \{1, \ldots, 128\}$ with $H < W$, on each of 11 traces: **7 recorded channels from two public datasets** — Sleep-EDF (Fpz-Cz, Pz-Oz, EOG) and MIT-BIH polysomnographic (ECG, blood pressure, O2-A1 EEG, nasal respiration) — plus 4 synthetic processes (white noise, AR(1) ϕ=0.5, AR(1) ϕ=0.95, random walk). Two datasets, picked for spread in autocorrelation rather than for count.

No (W, H) **pair violated the bound, on any trace** (286 grid cells in total). Observed/bound median 1.000, mean 0.981; the exact formula matches observed inflation to a maximum relative error of 1.3×10^{-15}.

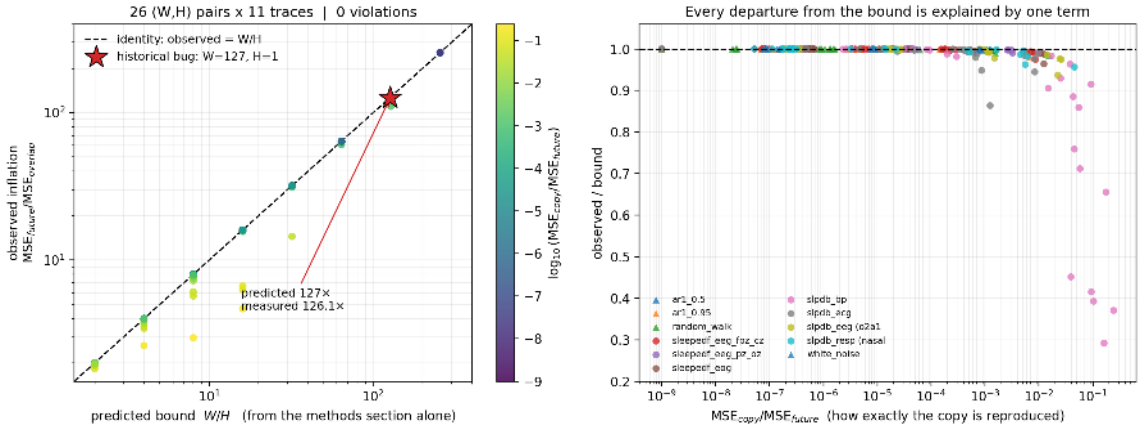


Figure 1: Left: predicted bound W/H against observed inflation, log–log, with the identity line. Points lie on it and never above it; the region where a counterexample would appear is empty. Colour is the correction term. The star is this project's own invalidated result (W=127, H=1): predicted 127×, measured 126.1×. **Right:** every departure from the bound is explained by the single term $\mathrm{MSE}_{\mathrm{copy}}/\mathrm{MSE}_{\mathrm{future}}$, monotonically, across five decades. Triangles are synthetic processes, circles real recordings.

Agreement is at machine precision because the relation is an identity; the code asserts the decomposition per cell, so an incorrect mask fails loudly rather than producing a plausible curve. Departures are monotone in the correction term: mean observed/bound is 1.000 below 10^{-5} (126 cells), 0.986 over 10^{-3}–10^{-2} (38 cells), 0.428 above 10^{-1} (4 cells). That monotonicity is the argument — the law is not approximately right by luck, it is exactly right with a known correction.

The sharpest test is white noise. A signal with no predictable structure still inflates at the bound (median 1.000): the copyable positions are in the input whether or not anything is forecastable. The artifact belongs to the *metric*, not to the signal or the model.

4 The corrected benchmark

Strictly-future scoring on real data: Sleep-EDF sleep-cassette, all 78 subjects, subject-disjoint 39 / 12 / 27, zero input–target overlap. Both models tuned per horizon on the same validation subjects, the ridge grid extended until no horizon selected a grid edge. Per-subject MSE, paired Wilcoxon, Holm-corrected across nine horizons.

h	ms	persistence	ridge-AR	GRU	Δ	Holm p
16	160	1.3814	0.8298	**0.7268**	**+0.1031**	$\mathbf{2.2 \times 10^{-6}}$
32	320	1.7600	**0.8700**	0.8756	−0.0056	n.s.
256	2560	1.8746	**0.9942**	1.0099	−0.0157	1.3×10^{-7}

At 160 ms the GRU is 12.4 % below a converged linear baseline on 26 of 27 held-out subjects and 47.4 % below persistence; the advantage ends by ∼320 ms. Eight falsification tests followed. Two matter most: on a synthetic AR(2) process, where the optimal predictor is provably linear, the same GRU with the same tuning budget **loses** to ridge (p=0.014), so the apparatus cannot manufacture wins; and a linear model trained by the GRU's exact optimiser is *worse* than closed-form ridge at every horizon, so the margin is architecture, not optimisation. **One claim did not survive:** we had reported tuned ridge beating the GRU at 640 ms (p=4.1×10^{-5}), but under non-overlapping windows that becomes a tie (p=0.58), so the claim is narrowed to "the advantage ends by 320 ms" rather than "linear wins beyond it".

Not claimed. No clinical utility — 160 ms is roughly 400× shorter than the ∼1 minute precursor timescale of interest. Single channel, one corpus. The law governs the *metric*, not model quality: a good model and a useless one inflate identically. The prevalence survey, establishing how often published work uses $H < W$, is not done; that is what would turn this from a tool into a finding about the field.

Artifacts. `artifacts/overlap_law/` and `artifacts/forecast_crossover/` hold the 286 cells, the figure, the benchmark and the eight-test validation; the corresponding write-ups are `docs/results/overlap_inflation_law_2026-07-28.md` and `forecast_crossover_2026-07-27.md`; produced by `scripts/overlap_inflation_law.py`, `forecast_scaleup_paired.py` and `validate_forecast_claim.py`. The invalidated 3.116 MSE / r=0.972 result is retained in `docs/results/a100-run-history.md` as evidence of the bug, not as performance.